

1. Title

House Price Prediction using Linear Regression

2.Objective

The objective of this project is to develop a machine learning model using Linear Regression to predict median house prices based on the California Housing dataset. The project includes data preprocessing, exploratory data analysis (EDA), model training, evaluation, and performance assessment using standard regression metrics such as MAE, RMSE, and R^2 score.

3.Dataset

The dataset used in this project is the California Housing Dataset, which is available in the scikit-learn library. It contains housing data collected from various districts in California and is widely used for regression-based machine learning tasks.

Source:

The dataset is provided by the sklearn.datasets module as a built-in dataset.

Features:

The dataset includes the following input features:

- Median Income (MedInc): Average income of households in the district
- House Age (HouseAge): Median age of houses in the district
- Average Rooms (AveRooms): Average number of rooms per household
- Average Bedrooms (AveBedrms): Average number of bedrooms per household
- Population: Total population of the district
- Average Occupancy (AveOccup): Average number of people per household
- Latitude: Geographic latitude of the district
- Longitude: Geographic longitude of the district

Target Variable:

- Median House Price (HousePrice): The target variable represents the median house value for each district, which is the value predicted by the machine learning model.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the structure and relationships within the dataset before building the machine learning model.

Correlation Analysis

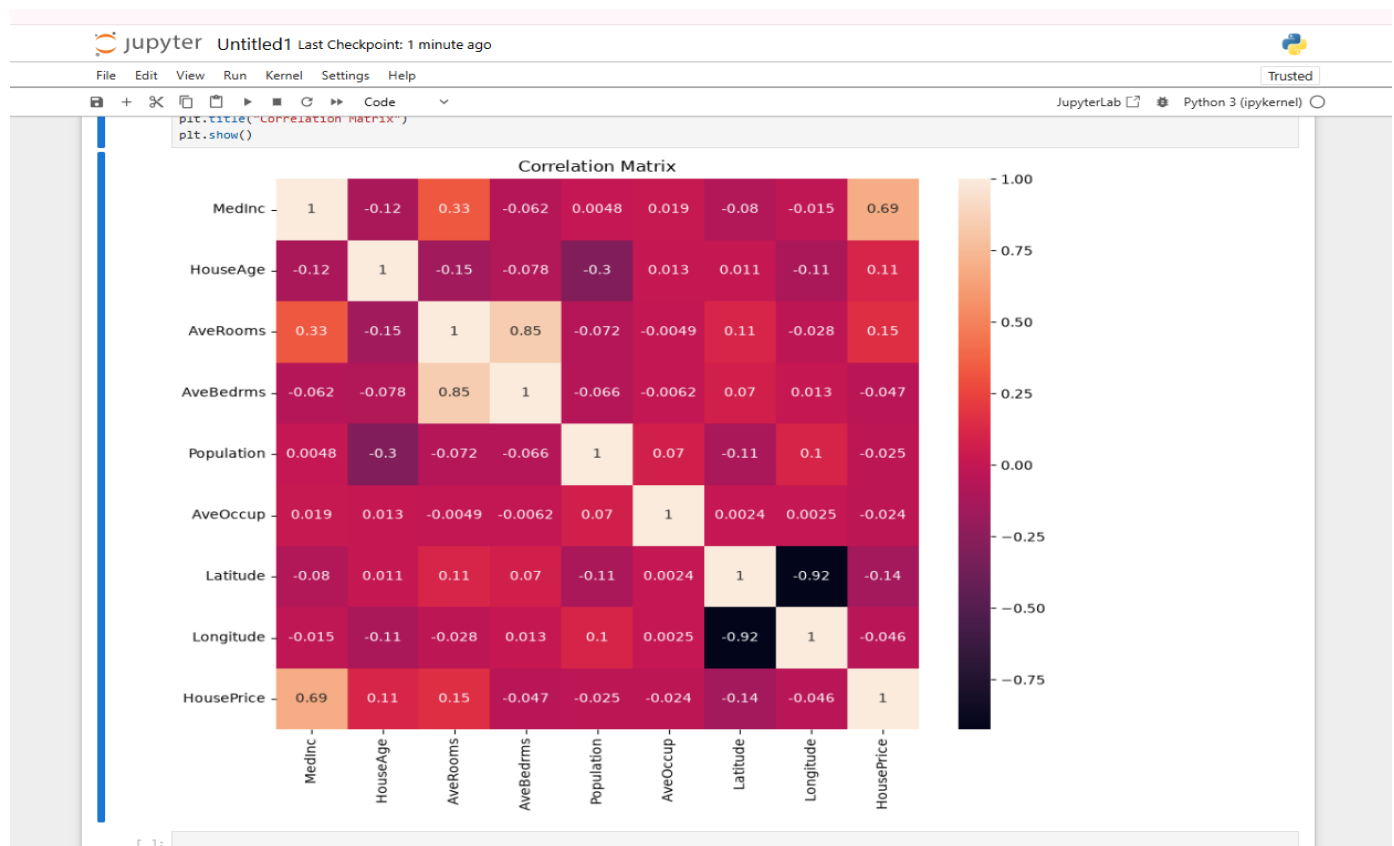
A correlation heatmap was generated to identify relationships between features and the target variable. It was observed that Median Income (MedInc) has the strongest positive correlation with the median house price. This indicates that income level is a major factor influencing housing prices in California.

Distribution Analysis

The distribution of the target variable (median house price) was visualized using a histogram. The results showed that the house prices are slightly right-skewed, meaning that most houses fall within a lower to moderate price range, with fewer high-priced houses.

Missing Values Check

The dataset was checked for missing values, and it was found that there are no missing values in any of the features. This made the dataset clean and suitable for model training without requiring imputation.



5. Model Used

The machine learning model used in this project is Linear Regression, which is a supervised learning algorithm used for predicting continuous numerical values. It works by establishing a linear relationship between the independent variables (features) and the dependent variable (target), which in this case is the median house price.

Algorithm:
Linear Regression

Train-Test Split:

The dataset was divided into training and testing sets using an 80/20 split ratio. This means 80% of the data was used to train the model and 20% of the data was used to evaluate its performance on unseen data. This approach helps in checking how well the model generalizes to new inputs.

Reason for Choosing Linear Regression:

Linear Regression was selected because it is a simple, efficient, and interpretable model for regression problems. It also serves as a strong baseline model for predicting continuous values such as house prices.

6. Evaluation Metrics

MAE: 0.5332001304956557

RMSE: 0.7455813830127758

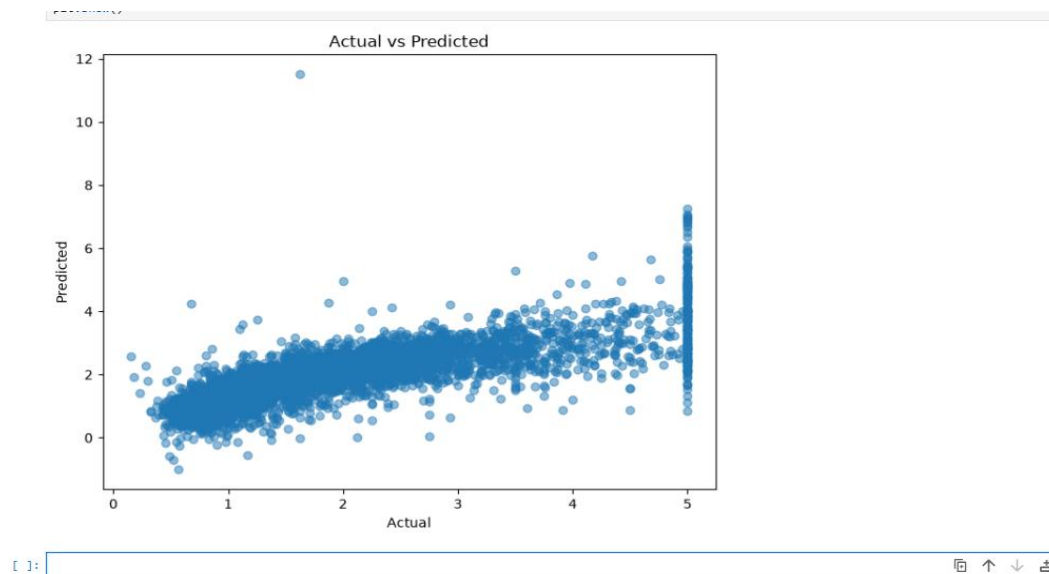
R2 Score: 0.5757877060324514

7. Results

The Linear Regression model achieved a moderate level of performance on the California Housing dataset. The model explains approximately 60% of the variance ($R^2 \approx 0.60$) in the target variable, indicating that it captures a significant portion of the relationship between the input features and house prices.

The predictions made by the model are reasonably accurate for a baseline regression model. However, there is still some deviation between the actual and predicted values, which is expected since Linear Regression assumes a simple linear relationship between variables.

Overall, the model performs satisfactorily as a baseline approach, and its performance can be further improved using more advanced algorithms such as Ridge Regression, Random Forest, or Gradient Boosting.



8. Conclusion

The Linear Regression model was successfully used to predict median house prices using the California Housing dataset and achieved moderate accuracy by capturing the overall trend in the data. However, some prediction errors were observed due to the simplicity of the model and the complexity of real-world data.

9. Future Improvements

The performance of the Linear Regression model can be improved by applying regularization techniques such as **Ridge and Lasso Regression** to reduce overfitting and improve generalization. Advanced models like **Random Forest** can also be used to capture non-linear relationships in the data. Additionally, applying **feature scaling** and performing **hyperparameter tuning** can further enhance model accuracy and overall performance.